

Assignment 2: Inverted Pendulum Walker — Report

Sketch

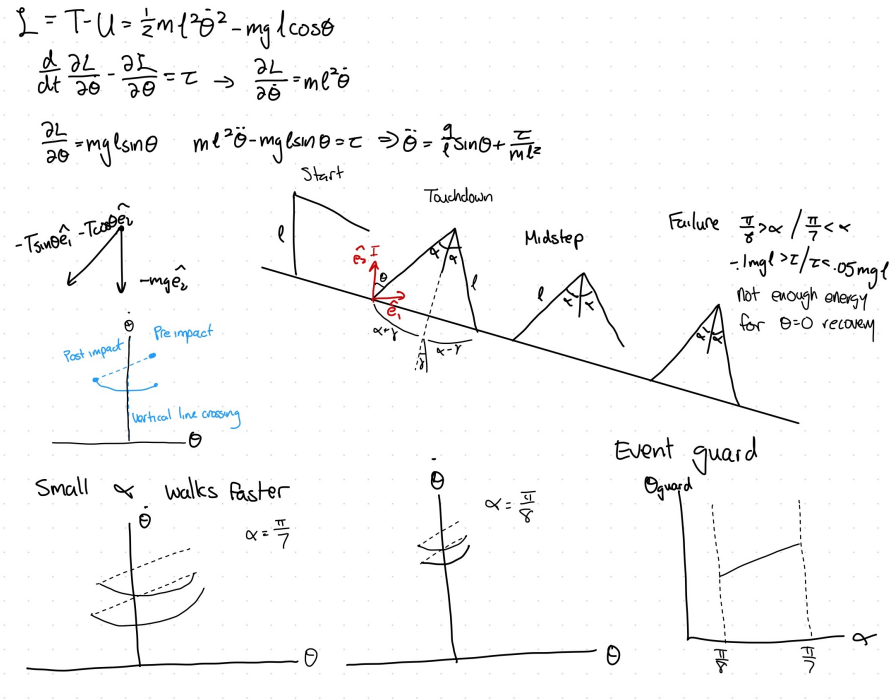


Figure 1: Model Sketch

Region of Attraction

The ankle controller combines feedback linearization (to cancel the gravity term, $g/l * \sin(\theta)$) with a PD stabilizing term:

torque = $m * l^2 * (-k_p \theta - k_d \dot{\theta} - (g/l) * \sin(\theta))$, clipped to $[-0.1 m g l, 0.05 m g l]$

with gains $k_p = 9$, $k_d = 6$ (chosen by trial and error for a smooth but not overly damped response). The region of attraction was found by a grid search over θ and $\dot{\theta}$ in $[-0.5, 0.5]$ (kept small and focused, per the assignment's guidance), simulating each starting point for 5 seconds and checking convergence to the origin:

The RoA forms a diagonal band containing 143 of the 400 grid points tested. It is not a symmetric disc around the origin: states where θ and $\dot{\theta}$

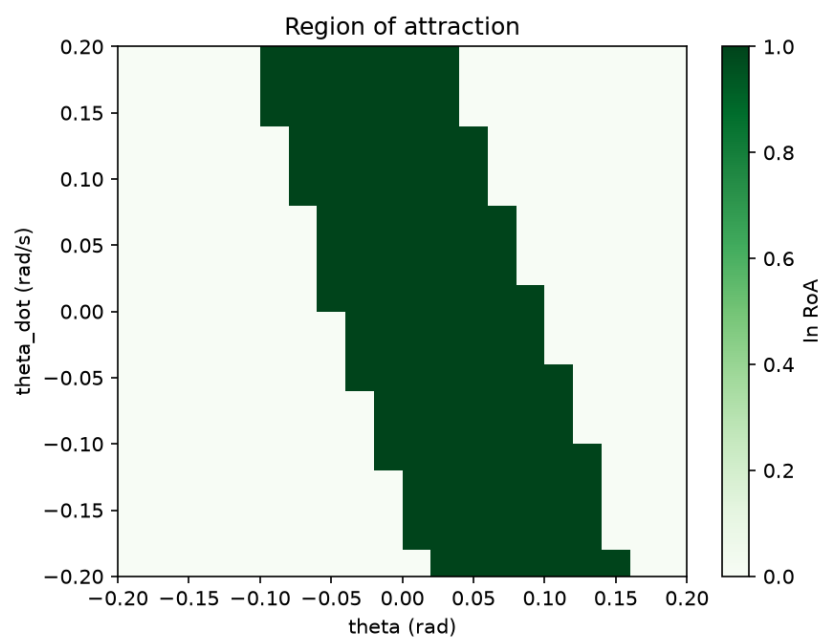


Figure 2: Region of attraction

have opposite signs (i.e. already moving back toward vertical) are captured over a wider range than states where both push in the same direction, which is consistent with the PD law's “least correction needed” line, roughly $\theta = -(k_d/k_p) * \dot{\theta}$.

Choice of Poincaré Section

In the rimless wheel, touchdown was a convenient section because the touchdown angle ($\theta_{TD} = \gamma + \alpha$) was a fixed constant (set by the spoke spacing). Here, α is a per-step control input rather than a geometric constant, so θ_{TD} moves every time a different α is chosen — touchdown is no longer a fixed value of θ , and using it as a section would violate the requirement that θ be constant on the section.

Instead, **$\theta = 0$ (mid-stance, the upright crossing)** is used as the section:

- **θ is constant on the section** — trivially, since it is defined by $\theta = 0$ regardless of what α was chosen for that step.
- **Transverse to the flow** — during normal forward walking, the leg swings through vertical with $\dot{\theta}$ not equal to 0 (it is mid-swing, not at a turning point), so trajectories cross the section rather than run tangent to it.
- **Leaves exactly one free state, $\dot{\theta}_k$** — at the crossing, $\theta = 0$ is fixed, so the only remaining coordinate is $\dot{\theta}$, which is exactly the single state the step-to-step return map needs.

The phase portrait below confirms this empirically: every pass through $\theta = 0$ during the sample trajectory has clearly nonzero slope (visible “corners” rather than tangencies), and the section correctly reduces the full continuous trajectory to a simple 1-D sequence of $\dot{\theta}_k$ values:

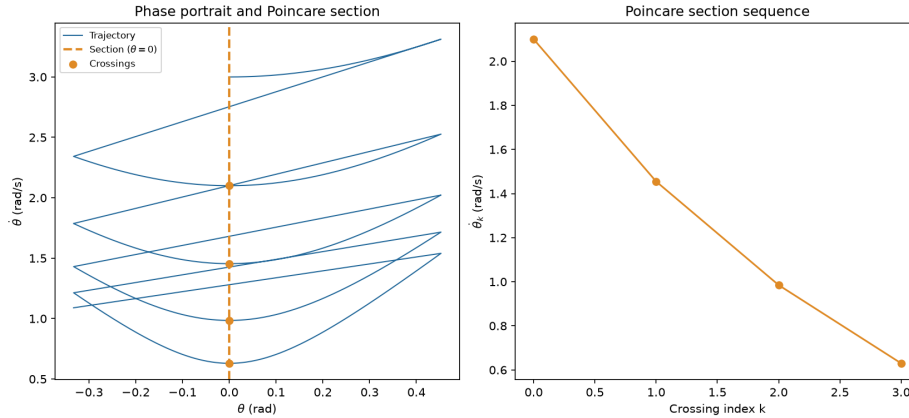


Figure 3: Poincaré section

Grid Resolution Verification

Criterion used: build the state-action lookup table at increasing resolutions (M , K), and at each resolution, actually simulate (full continuous dynamics, not just table lookups) how many real footsteps four test initial velocities ($\text{theta_dot_0} = 0.5, 1.5, 2.5, 3.5$ rad/s) take to reach the RoA, following that table’s own policy. These are compared against a much finer 35×21 reference table. A resolution is considered “fine enough” if every test point’s simulated step count agrees with the reference within 1 step (some jitter is expected from nearest-grid-point policy lookup even as resolution improves, so exact agreement at every resolution is not a meaningful bar).

M	K	$\text{theta_dot_0} = 0.5$	1.5	2.5	3.5	Result
35 (reference)	21	1	2	4	5	—
6	4	unreachable	unreachable	unreachable	unreachable	FAILS
9	6	1	unreachable	unreachable	4	FAILS
12	7	1	3	4	5	PASSES
15	9	1	3	3	4	passes
20	12	1	3	4	4	passes
25	15	1	2	4	4	passes

$M = 12$, $K = 7$ is the coarsest resolution that passes. The next-coarser candidate, $M = 9$, $K = 7$, fails outright: two of the four test velocities ($\text{theta_dot_0} = 1.5$ and 2.5) become entirely unreachable within the step cap at that resolution, even though the finer reference finds valid solutions (2 and 4 steps respectively) for both. $M = 6$ fails even more severely, with every test point unreachable. This confirms $M = 12$ is both necessary ($M = 9$ is measurably worse) and sufficient ($M = 15$ and above give no further qualitative improvement, just minor jitter of ± 1 step).

At Least 3-Step Trajectory and Maximum Sustainable Steps

Using the chosen $M = 12$, $K = 7$ table, $\text{theta_dot_0} = 4.429$ rad/s (the Froude-2 bound itself) is used as the example initial condition. This point was chosen deliberately: it is one of several grid states where the optimal (fewest-footsteps) and maximally-delaying policies select a **different** α on the very first footstep, so the two trajectories visibly diverge from the start rather than tracking identically for a few steps before splitting apart.

The table predicts the optimal policy needs 4 steps at this resolution; simulating the real continuous trajectory under that policy actually takes **6 real footsteps** to reach the RoA (the discrepancy between the table’s prediction and the real simulation is the expected discretization error at this resolution, consistent with the ± 1 -step tolerance established above — here it compounds slightly further since several steps are chained together before reaching the RoA).

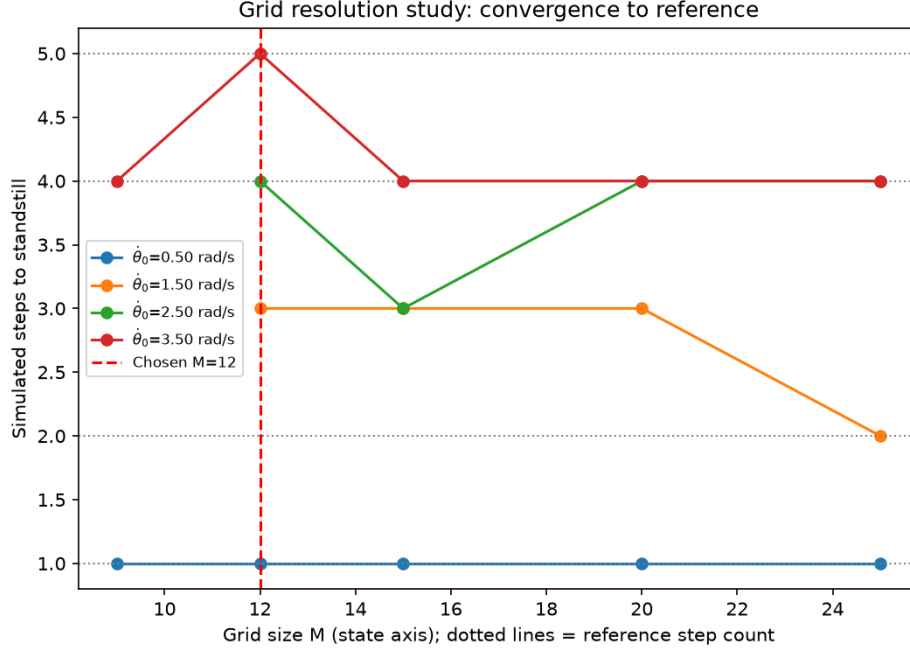


Figure 4: Grid resolution study

For this same initial condition, the maximum number of steps the walker could be made to sustain before being forced into the RoA — using a deliberately delaying (but still valid) choice of α at each step, rather than the fastest route in — is **7 steps**.

The left panel shows θ over time for both policies, with footsteps marked: the thick, semi-transparent orange trace is the maximally-delaying policy, and the dashed blue trace on top of it is the optimal policy. The two diverge immediately (different footstrike timing from the very first step), and the optimal trajectory reaches the RoA and stops one footstrike earlier than the delaying one. The right panel summarizes the two step counts (6 vs. 7) side by side.

Steps to Standstill by Initial Condition

For every state on the chosen $M = 12$ grid, the minimum number of footsteps to reach standstill (using the best available α at each step) was computed by backward chaining from the states that reach the RoA directly:

$\theta_{\dot{k}}$ (rad/s)	Best α (rad)	Steps to standstill
0.000	—	unreachable
0.403	0.393	1
0.805	0.421	1

theta_dot_k (rad/s)	Best alpha (rad)	Steps to standstill
1.208	0.393	2
1.611	0.411	2
2.013	0.393	3
2.416	0.393	3
2.819	0.421	3
3.221	0.393	4
3.624	0.393	4
4.027	0.393	4
4.429	0.411	4

Only $\text{theta_dot_k} = 0$ is unreachable, which is expected: it is an exact fixed point of the passive swing dynamics ($\text{theta_ddot} = g/l * \sin(0) = 0$), so a footstep never actually gets taken from rest without some external push. Step count grows monotonically with initial velocity, from 1 step near theta_dot_k of about 0.4-0.8 up to 4 steps by the Froude-2 bound (theta_dot_k of about 4.43) — consistent with the intuition that more initial energy requires more footsteps to shed before the ankle controller can take over.

The left panel is the state-action lookup table itself: the dark region shows which $(\text{theta_dot_k}, \alpha)$ pairs land directly in the RoA in a single footstep. The right panel is the resulting steps-to-standstill curve described above.

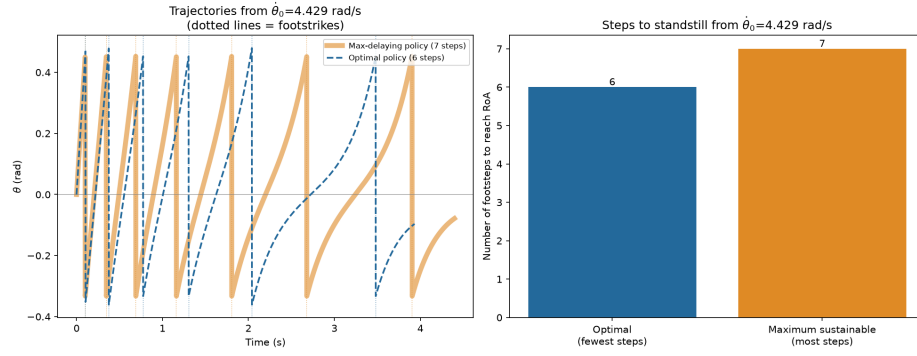


Figure 5: At least 3-step trajectory and maximum sustainable steps

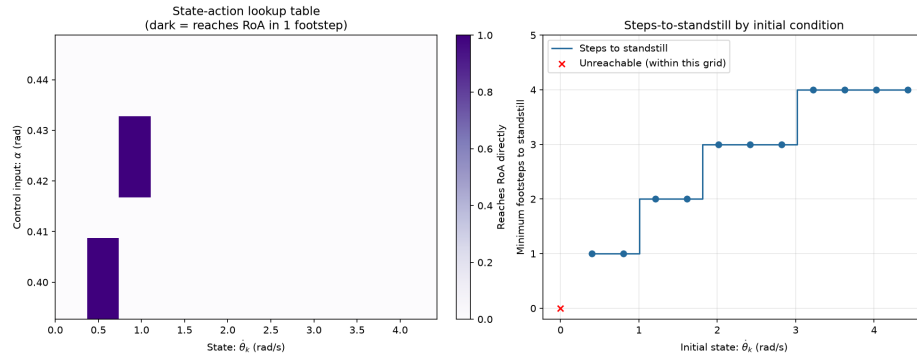


Figure 6: State-action table and steps-to-standstill